

# Financial Controlling Dashboard

Excel-based financial controlling model with Power Pivot for EPC/construction project portfolio. Features EAC/ETC variance analysis, WIP tracking, cash forecasting, and commitment management.

Excel

Power Pivot

Power Query

DAX

Star Schema

## Project Overview

This project is a financial controlling model for a fictional EPC contractor called **Northbridge Engineering & Construction Ltd.** The company manages 66 industrial projects across USA regions.

The goal is to build a complete controlling model that shows real-world CFO/Controller reporting:

- Portfolio performance monitoring
- Cost variance and EAC analysis
- Cash flow and commitment tracking

## Key Metrics

BAC (BUDGET)

**\$916M**

EAC (ESTIMATE)

**\$821M**

ETC (TO COMPLETE)

**\$13.8M**

EAC VS BAC

**-\$95.2M**

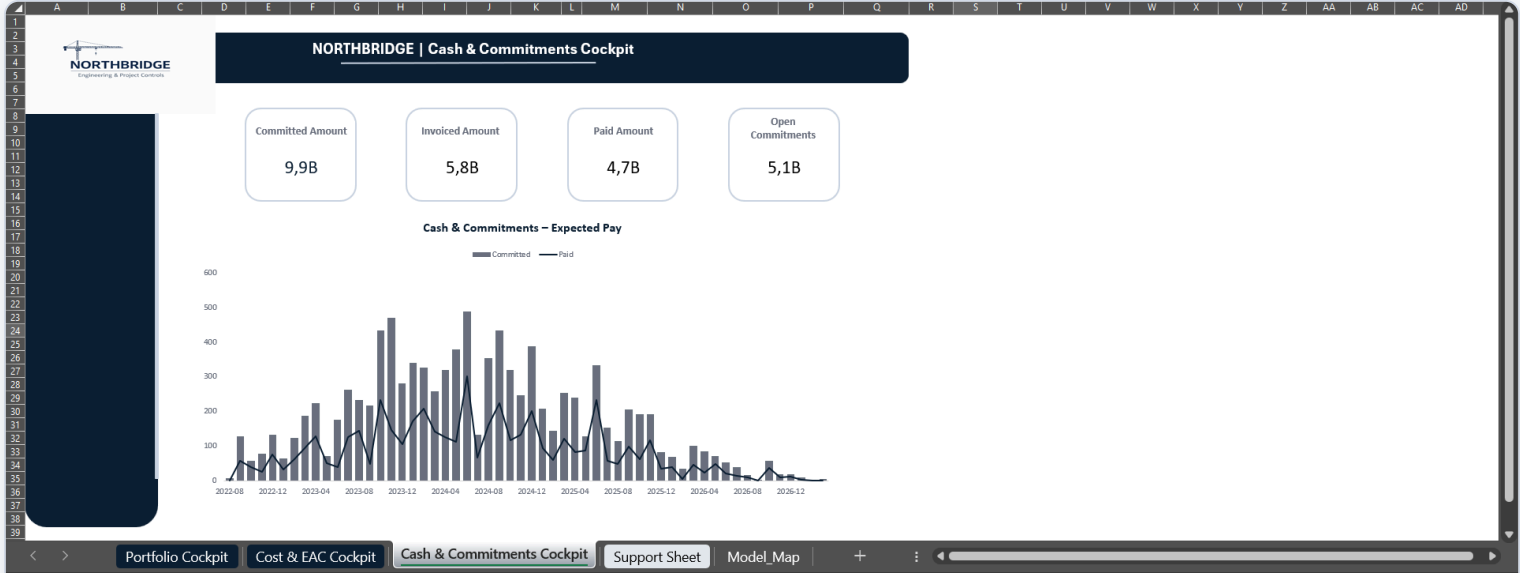
COMMITMENTS

**\$9.9B**

OPEN COMMITS

**\$5.1B**

## Dashboard Screenshots



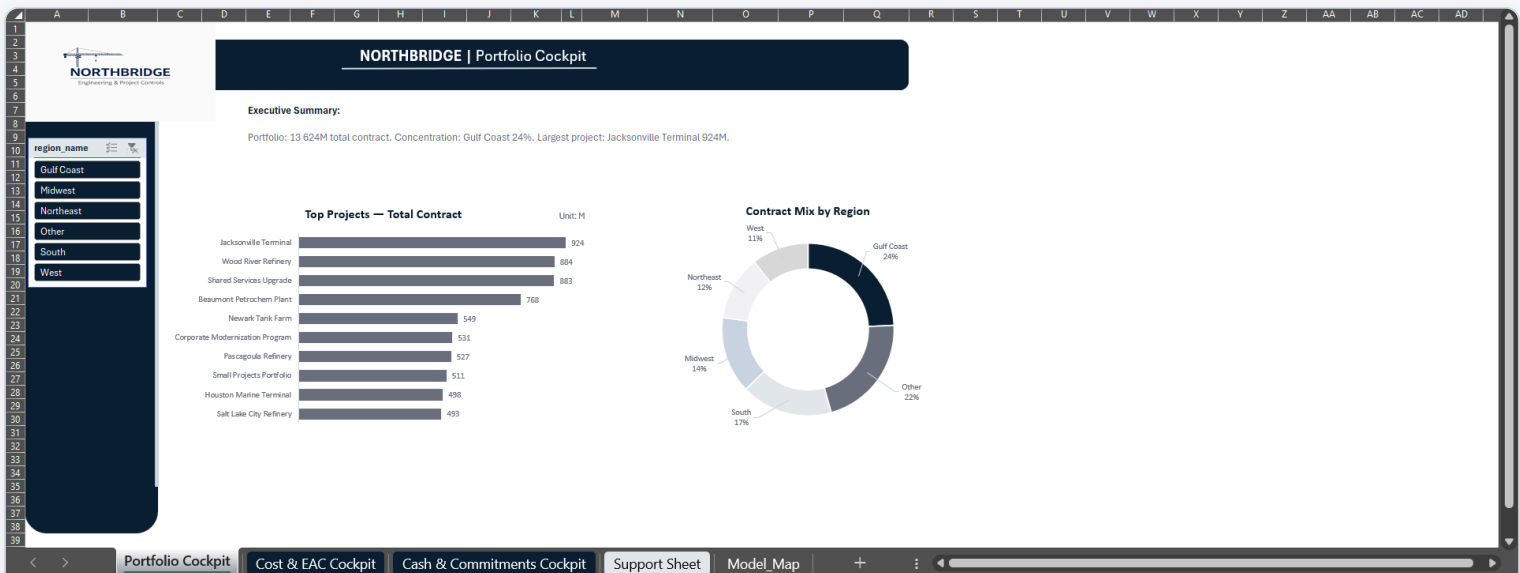
## Portfolio Cockpit

Executive overview with Top Projects ranking and regional distribution chart.



## Cost & EAC Cockpit

Variance analysis (EAC vs BAC), cash forecast with snapshot month selection.

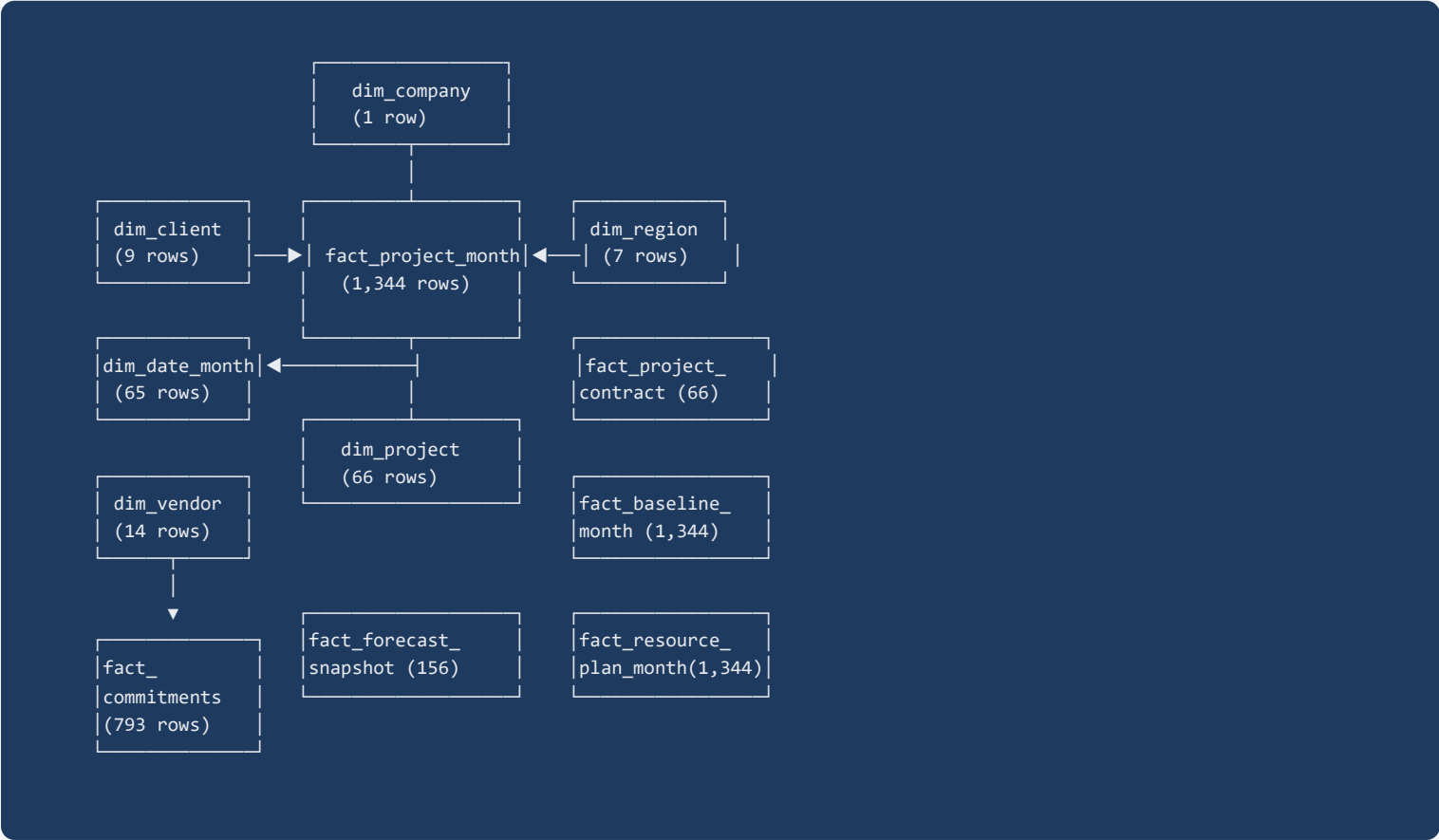


## Cash & Commitments Cockpit

Commitment lifecycle tracking: Committed → Invoiced → Paid amounts over time.

## Data Model (Star Schema)

The model uses a **star schema** architecture. This is the best practice for Power Pivot analysis. It has 6 dimension tables and 6 fact tables.



### Dimension Tables

| TABLE          | ROWS | DESCRIPTION                                 |
|----------------|------|---------------------------------------------|
| dim_company    | 1    | Company master (Northbridge Engineering)    |
| dim_client     | 9    | Client / customer master                    |
| dim_region     | 7    | USA regions (Gulf Coast, West, South, etc.) |
| dim_project    | 66   | Project master with contract details        |
| dim_vendor     | 14   | Vendor / subcontractor master               |
| dim_date_month | 65   | Calendar dimension (2022-2027)              |

Fact Tables

| TABLE                    | ROWS  | GRAIN                      | DESCRIPTION                                   |
|--------------------------|-------|----------------------------|-----------------------------------------------|
| fact_project_month       | 1,344 | Project × Month            | Main fact: earned, invoiced, cash, costs, WIP |
| fact_project_contract    | 66    | Project                    | Contract values, budget, planned margin       |
| fact_baseline_month      | 1,344 | Project × Month            | Planned progress and value                    |
| fact_commitments         | 793   | PO Line                    | Purchase orders and payment tracking          |
| fact_forecast_snapshot   | 156   | Snapshot × Month × Project | Rolling EAC/cash forecasts                    |
| fact_resource_plan_month | 1,344 | Project × Month            | Labor hours and cost planning                 |

Key Measures (DAX)

Cost Performance

```
BAC = SUM(fact_project_contract[budget_total_cost_pln])
EAC = SUM(fact_forecast_snapshot[forecast_eac_cost])
ETC = [EAC] - [Actual Cost To Date]
EAC vs BAC = [EAC] - [BAC]
```

Revenue & Margin

```
Earned Revenue = SUM(fact_project_month[earned_revenue_pln])
Invoiced Revenue = SUM(fact_project_month[invoiced_revenue_pln])
Cash Received = SUM(fact_project_month[cash_received_pln])
WIP = [Earned Revenue] - [Invoiced Revenue]
```

Commitments

```
Committed Amount = SUM(fact_commitments[committed_amount])
Invoiced Amount = SUM(fact_commitments[invoiced_amount])
Paid Amount = SUM(fact_commitments[paid_amount])
Open Commitments = [Committed Amount] - [Paid Amount]
```

# Technical Stack

| Component     | Technology                             |
|---------------|----------------------------------------|
| Data Model    | Star Schema (6 dimensions + 6 facts)   |
| ETL           | Power Query (folder import)            |
| Analysis      | Power Pivot (Data Model)               |
| Visualization | Excel PivotCharts + Slicers            |
| Dashboards    | 3 Cockpits (Portfolio, Cost/EAC, Cash) |

## Skills Demonstrated

-  Star schema data modeling
-  Power Query ETL (folder import)
-  Power Pivot relationships
-  DAX measures
-  Financial KPIs (EAC, ETC, WIP)
-  Executive dashboard design
-  Variance analysis
-  Cash flow forecasting



View on GitHub

Portfolio project for Financial Controller / Data Analyst role